Worksheet: Strings in Python

Section A: MCQs (1 Mark Each)

1. What is the output of the following code?

   ```
   s = "Python"
   print(s[1:4])
   ```
   a) Pyt b) yth c) tho d) ytho
   Answer: b) yth
2. Which of the following methods is used to remove any whitespace from the beginning or the end of a string?
   a) strip() b) split() c) replace() d) find()
   Answer: a) strip()

3. What will be the output of the following code?

   ```
   s = "hello"
   print(s.upper())
   ```
   a) HELLO b) hello c) error d) hELLO
   Answer: a) HELLO
4. Which of the following is not a valid string method in Python?
   a) upper() b) reverse() c) split() d) join()
   Answer: b) reverse()

5. What is the result of the following expression?

   ```
   "Python" * 2
   ```

   a) Python2 b) Python Python c) PythonPython d) Error

   Answer: c) PythonPython

Section B: Very Short Questions (2 Marks Each)

1. **What is a string in Python?**
   Answer: A string in Python is a sequence of characters enclosed within single quotes (' ') or double quotes (" "). Strings are immutable, meaning once defined, their contents cannot be changed.
2. **How can you access the last character of a string?**
   Answer: You can access the last character of a string using negative indexing. For example, if s = "Python", s[-1] will return "n".
3. **Explain the use of the find() method.**
   Answer: The find() method returns the lowest index of the substring if it is found in the string. If not, it returns -1. Example: "hello".find("e") will return 1.
4. **What does the split() method do in a string?**
   Answer: The split() method breaks up a string into a list based on a specified delimiter. By default, the delimiter is whitespace. For example, "Hello World".split() will return ['Hello', 'World'].

5. **What is string slicing in Python? Give an example.**
   Answer: String slicing is a way to extract a part of a string. The syntax is string[start:end], where start is inclusive and end is exclusive. Example: "Python"[1:4] returns "yth".

Section C: Programming Questions (3 Marks Each)

1. **Write a Python program to check if a string is a palindrome.**
   Answer:

```
def is_palindrome(s):
    s = s.lower()
    return s == s[::-1]

# Test the function
string = "Madam"
if is_palindrome(string):
    print(f"{string} is a palindrome")
else:
    print(f"{string} is not a palindrome")
```

2. **Write a Python program to count the number of vowels in a given string.**
   Answer:

```
def count_vowels(s):
    vowels = "aeiouAEIOU"
    count = 0
    for char in s:
        if char in vowels:
            count += 1
    return count

# Test the function
string = "Hello World"
print(f"Number of vowels: {count_vowels(string)}")
```

3. **Write a Python program to reverse a given string.**
   Answer:

```
def reverse_string(s):
    return s[::-1]


# Test the function
string = "Python"
print(f"Reversed string: {reverse_string(string)}")
```

4. **Write a Python program to replace all spaces in a string with an underscore (_).**
   Answer:

```
def replace_spaces(s):
    return s.replace(" ", "_")

# Test the function
string = "Hello World"
print(f"Modified string: {replace_spaces(string)}")
```

5. **Write a Python program to find the length of a string without using the len() function.**
   Answer:

```
def string_length(s):
    count = 0
    for char in s:
        count += 1
    return count

# Test the function
string = "Hello"
print(f"Length of the string: {string_length(string)}")
```

Worksheet: Strings in Python (Part 2)

Section A: MCQs (1 Mark Each)

1. What will be the output of the following code?

```
s = "Python Programming"
print(s[::2])
```

   a) Pto rgamn b) Pto rgamnin c) Pto rgamnig d) PthnPormig

   Answer: d) PthnPormig

2. Which of the following expressions will return the string "Hello"?
   a) ''.join(['H', 'e', 'l', 'l', 'o'])
   b) '+'.join(['H', 'e', 'l', 'l', 'o'])
   c) '-'.join(['H', 'e', 'l', 'l', 'o'])
   d) ' '.join(['H', 'e', 'l', 'l', 'o'])

   Answer: a) ''.join(['H', 'e', 'l', 'l', 'o'])

3. What does the rfind() method do?

   a) Finds the first occurrence of a substring.
   b) Finds the last occurrence of a substring.

c) Replaces all occurrences of a substring.
d) Replaces the first occurrence of a substring.

Answer: b) Finds the last occurrence of a substring.

4. What will be the output of the following code?
   s = "hello"
   print(s.capitalize())
   a) HELLO b) Hello c) hellO d) hello
   Answer: b) Hello
5. Which method would you use to check if a string starts with a particular substring?
   a) startswith() b) endswith() c) find() d) index()
   Answer: a) startswith()

Section B: Very Short Questions (2 Marks Each)

1. **What is string immutability?**
   Answer: String immutability means that once a string is created, it cannot be changed. Any operation that alters the string creates a new string object instead of modifying the original one.
2. **How do you concatenate two strings in Python?**
   Answer: Strings can be concatenated using the + operator. For example, "Hello" + " World" results in "Hello World".
3. **Explain the use of the replace() method in strings.**
   Answer: The replace() method returns a copy of the string where all occurrences of a substring are replaced with another substring. Example: "apple".replace("a", "o") results in "opple".
4. **How do you find the length of a string in Python?**
   Answer: The len() function is used to find the length of a string. Example: len("Python") returns 6.
5. **What is the difference between isupper() and upper() in Python?**
   Answer: The isupper() method checks if all the characters in a string are uppercase and returns a boolean. The upper() method converts all the characters in a string to uppercase and returns a new string.

Section C: Programming Questions (3 Marks Each)

1. **Write a Python program to remove vowels from a given string.**
   Answer:

```
def remove_vowels(s):
    vowels = "aeiouAEIOU"
    result = ""
    for char in s:
        if char not in vowels:
            result += char
    return result
# Test the function
string = "Programming is fun"
print(f"String without vowels: {remove_vowels(string)}")
```

2. **Write a Python program to count the number of occurrences of a substring in a given string.**
   Answer:

```
def count_substring(s, sub):
    return s.count(sub)

# Test the function
string = "Hello, Hello, how are you?"
substring = "Hello"
print(f"Occurrences of '{substring}': {count_substring(string, substring)}")
```

3. **Write a Python program to check if a given string starts with a vowel.**
   Answer:

```
def starts_with_vowel(s):
    vowels = "aeiouAEIOU"
    return s[0] in vowels

# Test the function
string = "Apple"
if starts_with_vowel(string):
    print(f"{string} starts with a vowel")
else:
    print(f"{string} does not start with a vowel")
```

4. **Write a Python program to find the longest word in a given sentence.**
   Answer:

```
def longest_word(sentence):
    words = sentence.split()
    return max(words, key=len)

# Test the function
sentence = "Python programming is interesting"
print(f"Longest word: {longest_word(sentence)}")
```

5. **Write a Python program to convert a given string to title case (each word starts with a capital letter).**
   Answer:

```
def to_title_case(s):
    return s.title()

# Test the function
string = "python programming language"
print(f"Title case: {to_title_case(string)}")
```

Worksheet: Strings in Python (Part 3)

Section A: MCQs (1 Mark Each)

1. What is the output of the following code?

   ```
   s = "Python"
   print(s[-6])
   ```

   a) n b) P c) y d) h

   Answer: b) P

2. Which method is used to return a copy of the string with leading and trailing whitespaces removed?
   a) rstrip() b) strip() c) lstrip() d) replace()
   Answer: b) strip()

3. What will be the output of the following code?

   ```
   s = "abcdef"
   print(s.index("e"))
   ```

   a) 3 b) 4 c) 5 d) Error

   Answer: b) 4

4. What is the result of the following expression?

   "Python" + "Rocks"

   a) PythonRocks b) Python Rocks c) Error d) None of the above

   Answer: a) PythonRocks

5. Which of the following will result in a string of all lowercase letters?

   a) s.lower() b) s.islower() c) s.upper() d) s.capitalize()

   Answer: a) s.lower()

Section B: Very Short Questions (2 Marks Each)

1. **How do you check if a string contains only numeric characters?**
   Answer: You can use the isnumeric() method. If the string contains only numeric characters, it returns True; otherwise, it returns False.

2. **Explain the join() method with an example.**
   Answer: The join() method is used to join elements of a sequence (like a list or tuple) into a single string, using a specified separator. Example: "-".join(['a', 'b', 'c']) results in "a-b-c".
3. **What is the difference between rindex() and rfind()?**
   Answer: Both methods return the highest index of the substring if found. The difference is that rfind() returns -1 if the substring is not found, whereas rindex() raises a ValueError.
4. **How do you compare two strings in Python?**
   Answer: Strings can be compared using comparison operators like ==, !=, <, >, <=, and >= based on lexicographical order (ASCII values).
5. **Explain string formatting using the format() method with an example.**
   Answer: The format() method allows for variable substitution and formatting. Example:

   "{} is {} years old".format("John", 25)

   Output: "John is 25 years old"

Section C: Programming Questions (3 Marks Each)

1. **Write a Python program to check if two strings are anagrams.**
   Answer:

   python
   Copy code
   def are_anagrams(str1, str2):
   return sorted(str1) == sorted(str2)

   # Test the function
   string1 = "listen"
   string2 = "silent"
   if are_anagrams(string1, string2):
   print(f"{string1} and {string2} are anagrams")
   else:
   print(f"{string1} and {string2} are not anagrams")

2. **Write a Python program to remove all special characters from a given string.**
   Answer:

   import re

   def remove_special_chars(s):
   return re.sub(r'[^A-Za-z0-9]+', '', s)

   # Test the function
   string = "Hello! How@ are# you?"
   print(f"String after removing special characters: {remove_special_chars(string)}")
3. **Write a Python program to check if a given string is a valid identifier.**
   Answer:
   def is_valid_identifier(s):

```
    return s.isidentifier()


# Test the function
string = "my_variable"
if is_valid_identifier(string):
    print(f"'{string}' is a valid identifier")
else:
    print(f"'{string}' is not a valid identifier")
```

4. **Write a Python program to print the frequency of each character in a string.**
   Answer:

```
def char_frequency(s):
    freq = {}
    for char in s:
        if char in freq:
            freq[char] += 1
        else:
            freq[char] = 1
    return freq

# Test the function
string = "hello world"
frequency = char_frequency(string)
print(f"Character frequencies: {frequency}")
```

5. **Write a Python program to reverse the words in a given sentence.**
   Answer:

```
def reverse_words(sentence):
    words = sentence.split()
    return ' '.join(reversed(words))

# Test the function
sentence = "Python is fun"
print(f"Reversed sentence: {reverse_words(sentence)}")
```